

SAUMIL SINGHAL

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SUMMARY

CS student specializing in AI/ML with hands-on experience in machine learning models, deepfake detection, and full-stack development. Skilled in cloud fundamentals, system design, and building scalable real-world solutions. Strong problem-solver with disciplined learning habits.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript **Web/Tech:** HTML, CSS, Node.js, Next.js, Flask **Database:** SQL
ML/Tools: NumPy, Scikit-learn, PyTorch, Sentence Transformers, AWS, Git, Tailwind CSS **Core CS:** DSA, OOP

PROJECTS

CommunityLearn – AI-Powered Community Learning Platform Jan 2025 – May 2025
React.js, Node.js, Flask, Sentence Transformers, Tailwind CSS

- Built a full-stack community learning platform with React frontend, Node.js backend, and a Flask-based AI microservice.
- Integrated `all-MiniLM-L6-v2` sentence transformer for semantic content similarity and personalized recommendations.
- Developed REST APIs for content embedding, semantic search, and keyword extraction to power intelligent content discovery.
- Implemented CI/CD pipeline via GitHub Actions with automated deployment for continuous delivery.

Career Quest – Personalised Course & Career Recommender Oct 2023 – Jan 2024
Python, Flask, Scikit-learn

- Built an ML-based recommender system mapping 50+ user interest profiles to curated career paths and Coursera courses.
- Applied K-Means clustering and cosine similarity ranking to generate personalized learning paths with ~85% relevance accuracy.
- Designed a responsive UI and integrated Flask REST APIs delivering dynamic recommendations with under 300ms response time.

TruthLens – Deepfake Image Detection Web App Jul 2024 – Sep 2024
Next.js, PyTorch, Hugging Face, Tailwind CSS

- Developed deepfake detection system using a fine-tuned ViT model (`vit-base-patch16-224-in21k`).
- Implemented Grad-CAM highlights to visualize manipulated image regions for explainability.
- Built a responsive UI with secure API communication and preprocessing checks for reliable inference.

EDUCATION

Vellore Institute of Technology (VIT), Bhopal 2023–2027
B.Tech in Computer Science and Engineering (AI & ML) — **CGPA: 8.68**

Class 12 (ISC): 81%
Class 10 (ICSC): 87%

CERTIFICATIONS

- PHP and MySQL for Web Application Development
- Career Essentials in Generative AI – Microsoft & LinkedIn
- Cloud Computing – NPTEL (63%)
- Internet of Things – NPTEL (84%)
- Applied Machine Learning in Python – Coursera